



Is Learning From AI Flop-Shows A Good Idea?

This article highlights weak spots in artificial intelligence (AI) using case studies and guides us to focus on specific risks



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If you were to improve something, what is the first thing you would do? If you are thinking of assessing past performance, you are right. It is the most efficient and effective method of understanding improvement areas and fixes.

And, it should not be any different for artificial intelligence (AI) projects. Automation and primitive forms of AI have been in the trial and operation for the past many years. Some have worked as expected, while some have failed. I have chosen a few use-cases that have either failed or have not shown full benefits as expected.

These case studies highlight weak spots in AI and guide us to focus on specific risks. Of course, these use-cases are merely a drop in the ocean, but they represent a few common aspects.

Microsoft chatbot Tay on Twitter

On March 23, 2016, Microsoft released an artificially-intelligent chatbot called Tay. Named after the acronym Thinking About You, Tay was developed to help understand

how AI would interact with human users online.

Subsequently, Tay was programmed to ingest tweets and learn from them to communicate by responding and interacting with those users. Mainly, the target audience was American young adults. (I do not think the outcome would have been any different if the target audience was any different.) However, this attempted experiment lasted only 24 hours before Tay had to be taken offline for publishing extreme and offensive racist as well as sexist tweets.

Tay was a classic example of a system that was vulnerable to burning and pertinent issues in the data science world—garbage in, garbage out. It was developed to learn from active real-time conversations on Twitter, but it could not filter offensive inputs or bigoted comments in the process. Ultimately, Tay learned from user responses and reflected the same kind of emotion or thinking. Since most of the tweets were abusive, racist and sexist, Tay's responses followed the pattern.

The second attempt to release Tay by Microsoft did not go too well either and the bot was taken down soon after the second release. It has not been up online since then.

Of course, Microsoft did not explicitly program Tay for discrimination of any kind. It is safe to assume that its learning data did not have any discriminating characteristics either. However, feedback loop from which Tay was supposed to learn had a flaw.

Racial discrimination in Amazon's same-day delivery

In early 2016, Amazon rolled out same-day delivery to its Prime programme subscribers, but only for a select group of American cities and for a select group of neighbourhoods. These neighbourhoods were the ones where



Example of Microsoft chatbot Tay communicating with a user on Twitter